

```
import warnings
warnings.filterwarnings('ignore')
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
!gdown https://drive.google.com/uc?id=1Io_VRNN_euiDlQggX2IQNT0Xd3zWxe4X
```

```
Downloading...
From: https://drive.google.com/uc?id=1Io_VRNN_euiDlQggX2IQNT0Xd3zWxe4X
To: /content/aerofit_treadmill.csv
100% 7.28k/7.28k [00:00<00:00, 18.3MB/s]
```

```
df = pd.read_csv('aerofit_treadmill.csv')
```

Basic checks on data

```
#data glimps check
df.head()
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
#data information overview
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Product         180 non-null   object
1   Age             180 non-null   int64
2   Gender          180 non-null   object
3   Education       180 non-null   int64
4   MaritalStatus   180 non-null   object
5   Usage           180 non-null   int64
6   Fitness         180 non-null   int64
7   Income          180 non-null   int64
8   Miles           180 non-null   int64
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

```
# data shape check
df.shape
```

```
(180, 9)
```

```
# datatype check for all columns
df.dtypes
```

0

Product	object
Age	int64
Gender	object
Education	int64
MaritalStatus	object
Usage	int64
Fitness	int64
Income	int64
Miles	int64

dtype: object

#Checking overall data patterns
df.describe(include='all')

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles	
count	180	180.000000	180	180.000000	180	180.000000	180.000000	180.000000	180.000000	
unique	3	NaN	2	NaN	2	NaN	NaN	NaN	NaN	
top	KP281	NaN	Male	NaN	Partnered	NaN	NaN	NaN	NaN	
freq	80	NaN	104	NaN	107	NaN	NaN	NaN	NaN	
mean	NaN	28.788889	NaN	15.572222	NaN	3.455556	3.311111	53719.577778	103.194444	
std	NaN	6.943498	NaN	1.617055	NaN	1.084797	0.958869	16506.684226	51.863605	
min	NaN	18.000000	NaN	12.000000	NaN	2.000000	1.000000	29562.000000	21.000000	
25%	NaN	24.000000	NaN	14.000000	NaN	3.000000	3.000000	44058.750000	66.000000	
50%	NaN	26.000000	NaN	16.000000	NaN	3.000000	3.000000	50596.500000	94.000000	
75%	NaN	33.000000	NaN	16.000000	NaN	4.000000	4.000000	58668.000000	114.750000	
max	NaN	50.000000	NaN	21.000000	NaN	7.000000	5.000000	104581.000000	360.000000	

There are 3 products, with KP281 being the most purchased (80 out of 180 customers, ~44%).

Young Customer Base : Average age: 28.8 years Median age: 26 years 75% of customers are 33 years or younger

Male-Dominated Audience 104 males (58%) vs 76 females (42%)

Majority Are Partnered 107 customers (59%) are partnered

Moderate Product Usage Average usage: 3.46 times per week Most common usage: 3 times per week

Fitness Levels Are Mid-Range Average fitness rating: 3.31 Majority fall between 3 and 4

Middle-to-Upper Income Segment Average income: 53,720 Top 25 58,668 Max income: \$104,581

Moderate Activity Levels (Miles) Average miles: 103 miles Median: 94 miles Wide range: 21–360 miles

- Observations from basic data check :
- 1. Data is in good shape having no null values & no correction, imputation required.
 - 2. Data includes 3 types of unique products, 2 gender, 2 maritalstatuses.
 - 3. 6 string type columns and 3 numeric type columns.
 - 4. Product : KP281, Gender : Male, MaritalStatus : Partenered are top contributors to data.

Value counts and unique attributes

```
#Product Distribution
product_counts = round(df['Product'].value_counts(normalize=True)*100,2)
product_counts
```

proportion	
Product	
KP281	44.44
KP481	33.33
KP781	22.22

dtype: float64

```
#Gender Distribution
Gender_counts = round(df['Gender'].value_counts(normalize=True)*100,2)
Gender_counts
```

proportion	
Gender	
Male	57.78
Female	42.22

dtype: float64

```
#Marital Status Share
MaritalStatus_Count = round(df['MaritalStatus'].value_counts(normalize=True)*100,2)
MaritalStatus_Count
```

proportion	
MaritalStatus	
Partnered	59.44
Single	40.56

dtype: float64

```
product_summary = df.groupby('Product').agg(Avg_Age=('Age', 'mean'),Avg_Usage=('Usage', 'mean'),Avg_Fitness=('Fitness', 'mean'))
product_summary = product_summary.round(2)
product_summary
```

	Product	Avg_Age	Avg_Usage	Avg_Fitness	Avg_Income	Avg_Miles
0	KP281	28.55	3.09	2.96	46418.02	82.79
1	KP481	28.90	3.07	2.90	48973.65	87.93
2	KP781	29.10	4.78	4.62	75441.58	166.90

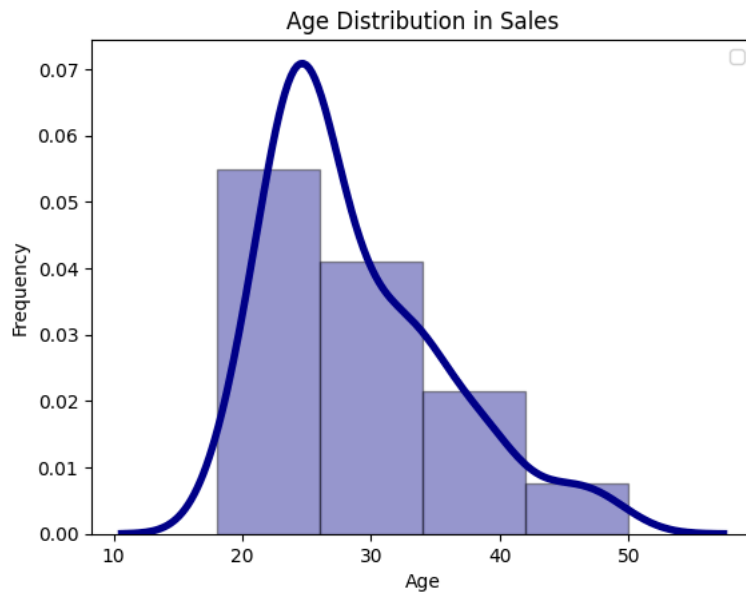
Next steps: [Generate code with product_summary](#) [New interactive sheet](#)

- Observations from value counts check :
- 1. Data includes 3 unique products with KP281 having highest contribution as 44% of total treadmill sales.
 - 2. Data is divided in Male and Female gender with Male having 15% higher contribution to product purchase.
 - 3. Partenered customers are puchasing trademills 48% more as compared to single customers.
 - 4. While age remains consistent across segments, KP781 attracts significantly higher-income, higher-fitness, and high-mileage customers, confirming its premium performance positioning.
 - 5. KP281 and KP481 serve the moderate-use segment with limited differentiation, suggesting potential repositioning opportunities.
 - 6. Customers tend to buy KP781 with high Avg_Usage, high Avg_Fitness high Avg_Income, high Avg_Miles

Visual Analysis

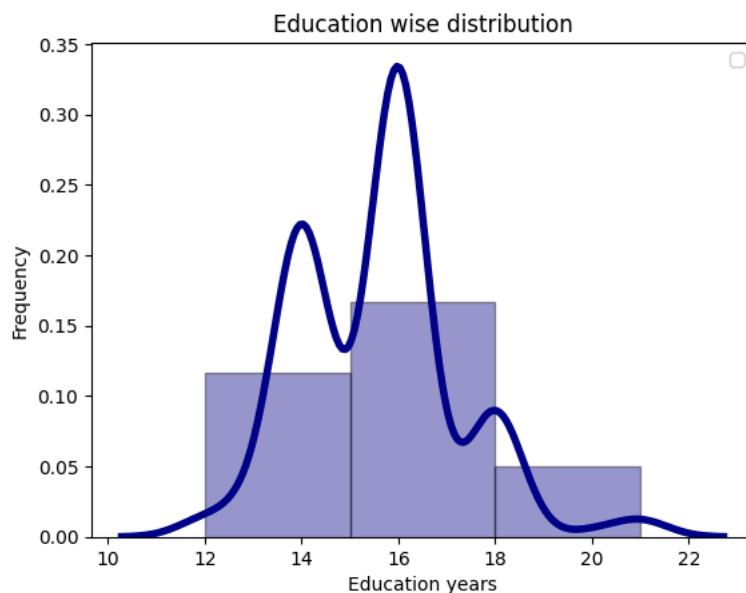
```
sns.distplot(df['Age'], hist=True, kde=True,bins=4, color = 'darkblue',hist_kws={'edgecolor':'black'})
,kde_kws={'linewidth': 4})
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.title('Age Distribution in Sales')
```

```
plt.legend()
plt.show()
```



1. Most customers fall between 20–35 years age group.
2. Peak Age Group is around 24–28 years age.
3. Ages 45–50 have very low frequency
4. The distribution appears right skewed as Mean>Median and Long tail toward higher ages
5. $P(20 \leq X \leq 35)$ is higher where $P(X > 40)$ is relatively small.

```
sns.distplot(df['Education'], hist=True, kde=True, bins=3, color = 'darkblue', hist_kws={'edgecolor':'black'},
kde_kws={'linewidth': 4})
plt.xlabel('Education years')
plt.ylabel('Frequency')
plt.title('Education wise distribution')
plt.legend()
plt.show()
```



1. Highest concentration around 16 years of education & noticeable group around 14 years
2. Very few customers with lower education below 13 levels.
3. Need to craft marketing messages aligned with educated, research-oriented buyers.

```
bins = [17, 25, 35, 45, 50]
labels = ['18-25', '26-35', '36-45', '46-50']
df['Age_Group'] = pd.cut(df['Age'], bins=bins, labels=labels, right=True)
df.head()
```

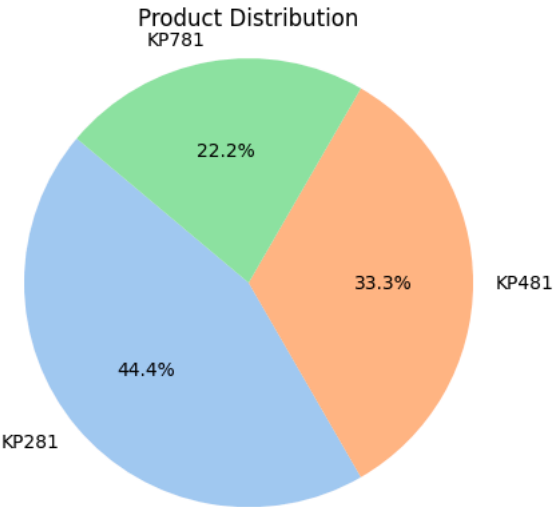
	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles	Age_Group	
0	KP281	18	Male	14	Single	3	4	29562	112	18-25	
1	KP281	19	Male	15	Single	2	3	31836	75	18-25	
2	KP281	19	Female	14	Partnered	4	3	30699	66	18-25	
3	KP281	19	Male	12	Single	3	3	32973	85	18-25	
4	KP281	20	Male	13	Partnered	4	2	35247	47	18-25	

Next steps:

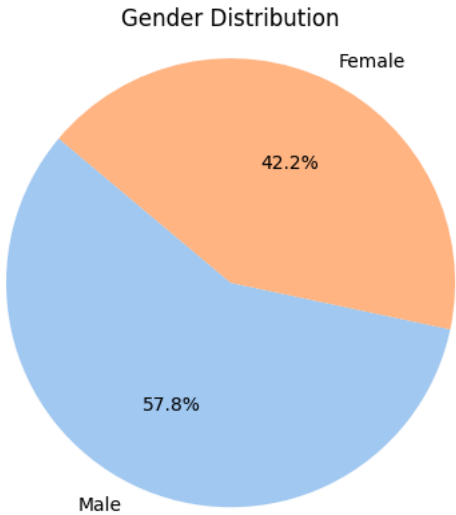
[Generate code with df](#)

[New interactive sheet](#)

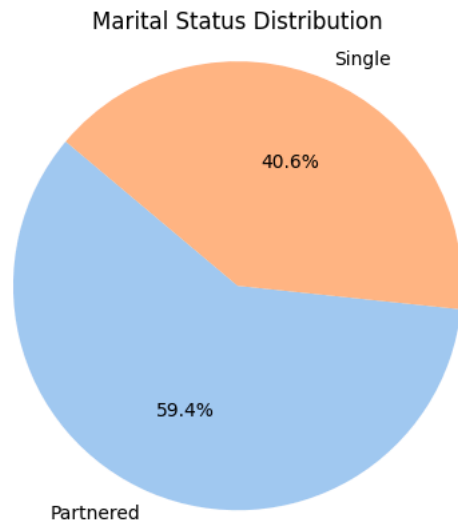
```
colors = sns.color_palette('pastel')[0:len(product_counts)]
plt.pie(product_counts, labels=product_counts.index, autopct='%1.1f%%', colors=colors, startangle=140)
plt.title('Product Distribution')
plt.axis('equal') # Equal aspect ratio ensures pie is circular.
plt.show()
```



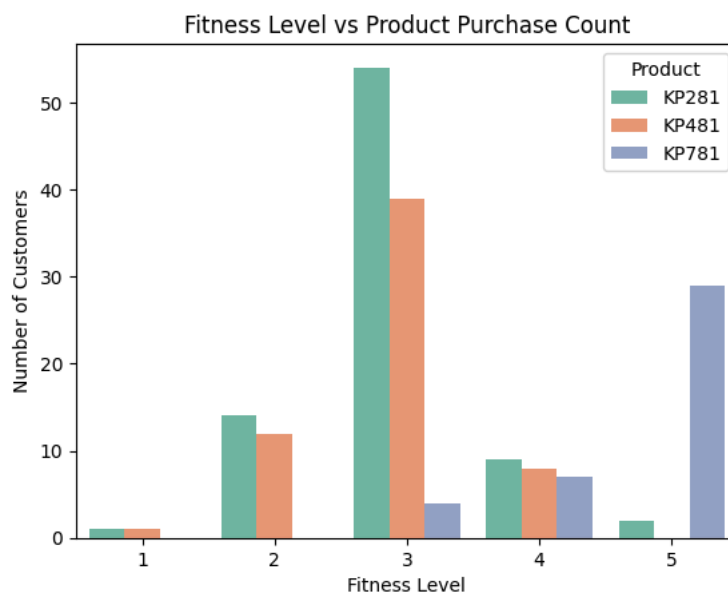
```
plt.pie(Gender_counts, labels=Gender_counts.index, autopct='%1.1f%%', colors=colors, startangle=140)
plt.title('Gender Distribution')
plt.axis('equal') # Equal aspect ratio ensures pie is circular.
plt.show()
```



```
plt.pie(MaritalStatus_Count, labels=MaritalStatus_Count.index, autopct='%1.1f%%', colors=colors, startangle=140)
plt.title('Marital Status Distribution')
plt.axis('equal') # Equal aspect ratio ensures pie is circular.
plt.show()
```

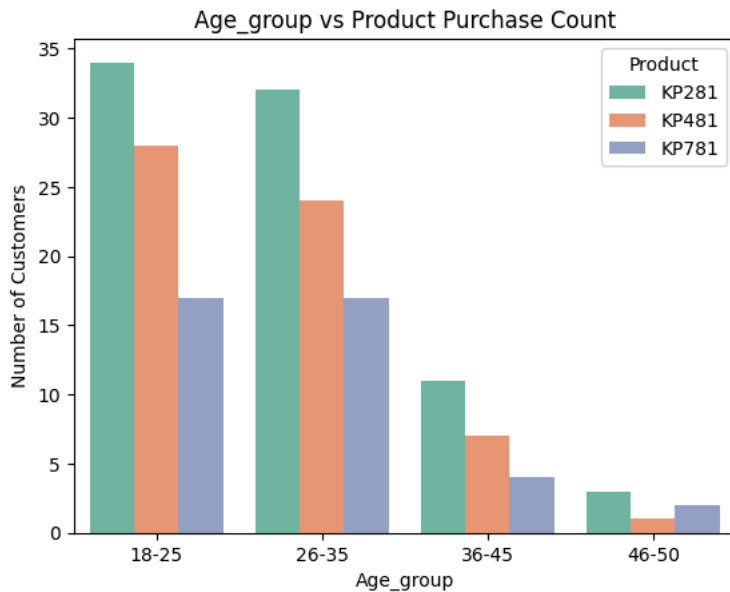


```
sns.countplot(data=df, x='Fitness', hue='Product', palette='Set2')
plt.title('Fitness Level vs Product Purchase Count')
plt.xlabel('Fitness Level')
plt.ylabel('Number of Customers')
plt.show()
```



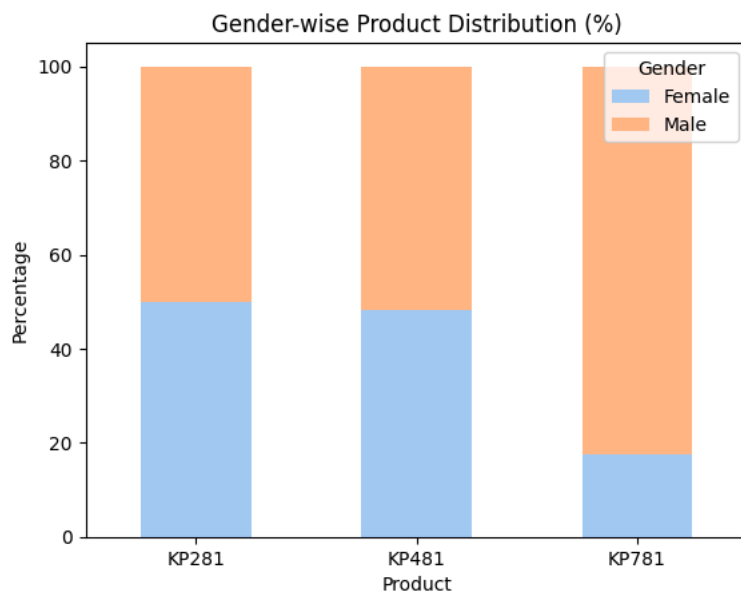
1. Most selling products KP281 and KP481 appeal mostly to customers with moderate fitness levels (2-4).
2. KP781 attracts primarily the most fit customers or highly active users.(Level 5).
3. The fitness level is a strong differentiator for product preference.
4. Marketing for KP781 should target highly fit individuals like athletes, fitness enthusiasts.
5. KP281 and KP481 should focus on moderately active customers.

```
sns.countplot(data=df, x='Age_Group', hue='Product', palette='Set2')
plt.title('Age_group vs Product Purchase Count')
plt.xlabel('Age_group')
plt.ylabel('Number of Customers')
plt.show()
```



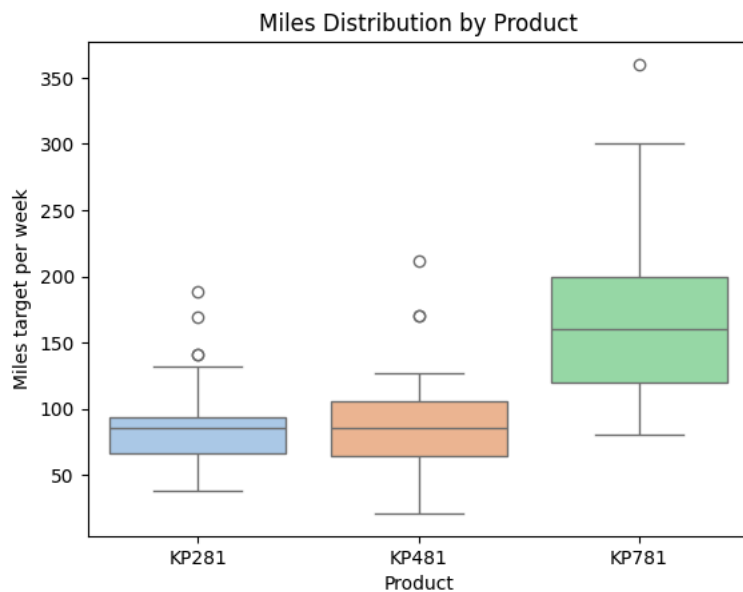
1. KP281 has the highest sales in every age group.
2. The 18–35 group is the largest buyer segment. They account for the majority of total sales across all products.
3. Sales decrease significantly in age group above 35.
4. Marketing strategies must be designed with focus on youth.
5. Offer loyalty programs to retain young buyers.
6. Social media marketing and Student discounts can be used.

```
gender_product_pct = pd.crosstab(df['Product'], df['Gender'], normalize='index') * 100
gender_product_pct.plot(kind='bar', stacked=True, color=colors)
plt.title('Gender-wise Product Distribution (%)')
plt.xlabel('Product')
plt.xticks(rotation=0)
plt.ylabel('Percentage')
plt.legend(title='Gender')
plt.show()
```



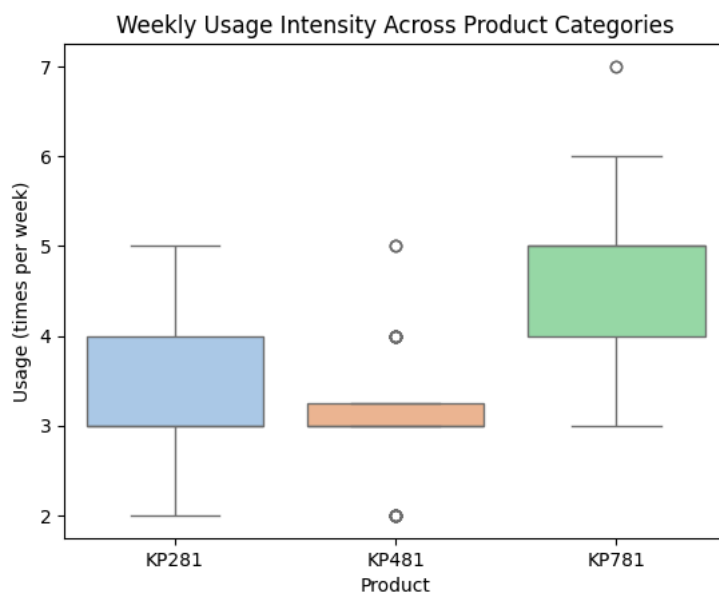
1. KP781 is a male-dominant premium product.
2. KP281 & KP481 are gender-neutral products.
3. KP781 has Potential to explore female-targeted campaigns or product tweaks.

```
sns.boxplot(data=df, x='Product', y='Miles', palette=colors)
plt.title('Miles Distribution by Product')
plt.xlabel('Product')
plt.ylabel('Miles target per week')
plt.show()
```



1. KP781 has the highest median miles, significantly above KP281 and KP481.
2. KP781 shows the widest spread (highest variability), indicating diverse but intensive usage patterns.
3. KP281 and KP481 have similar median miles, with heavy overlap in distributions.
4. KP481 shows slightly more variability than KP281 but not a major shift in central tendency.
5. KP781 has higher extreme values (outliers), suggesting a segment of very high-mileage users.
6. Position KP781 for serious runners with endurance-focused features and advanced tracking capabilities.
7. Differentiate KP481 more clearly, as its mileage distribution overlaps heavily with KP281.
8. Introduce tiered performance programs or add-ons for high-mile KP781 users to increase retention and upsell opportunities.

```
sns.boxplot(data=df, x='Product', y='Usage', palette=colors)
plt.title('Weekly Usage Intensity Across Product Categories')
plt.xlabel('Product')
plt.ylabel('Usage (times per week)')
plt.show()
```



1. KP781 has the highest weekly usage, with a median around 4–5 times and peaks up to 7 times per week.

2. KP281 shows moderate usage, typically between 3 - 4 times weekly with relatively stable variation.
3. KP481 has the lowest median usage (~3 times per week) and a tighter central distribution.
4. KP781 exhibits greater variability, indicating stronger engagement among certain user segments.
5. Upsell & bundle KP781 with complementary products to maximize revenue from highly engaged users.
6. Boost KP481 engagement through targeted promotions, user education, or feature enhancements to increase frequency of use.
7. Stabilize KP281 growth by implementing loyalty incentives to shift moderate users toward higher weekly usage.

```
zscore_df = product_summary.copy()

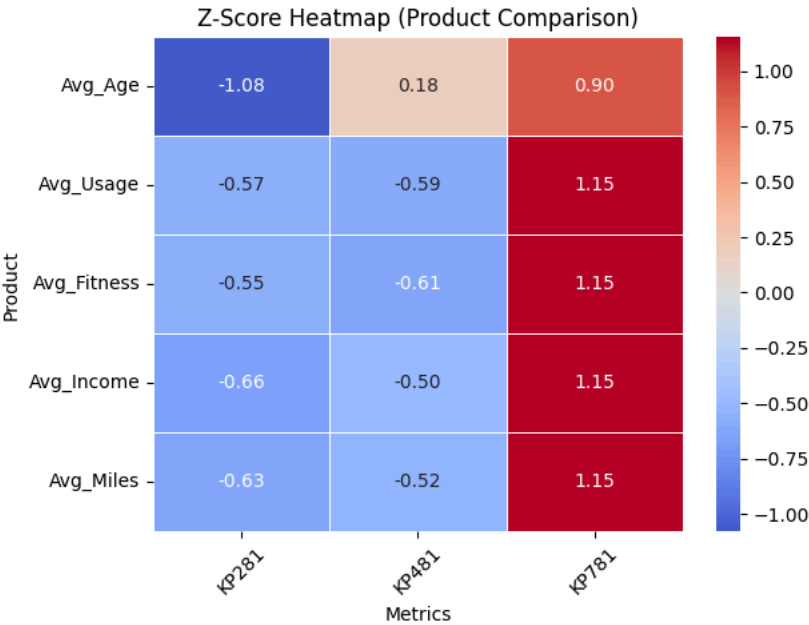
features = zscore_df.columns.drop("Product")

zscore_df[features] = (zscore_df[features] - zscore_df[features].mean()) / zscore_df[features].std()

zscore_df.set_index("Product", inplace=True)

sns.heatmap(zscore_df.T,annot=True,cmap="coolwarm",center=0,fmt=".2f",linewidths=0.5)

plt.title("Z-Score Heatmap (Product Comparison)")
plt.ylabel("Product")
plt.xlabel("Metrics")
plt.xticks(rotation=45)
plt.show()
```



1. KP781 strongly over-indexes on Usage, Fitness, Income, and Miles (~ +1.15), indicating a high-performance, high-income segment.
2. Age is not a major differentiator — differences are relatively small compared to other metrics.
3. KP281 underperforms across most metrics, especially Income and Engagement-related measures.
4. KP481 sits close to average but slightly below on performance indicators, showing weak differentiation.
5. Clear two-cluster structure emerges: KP781 (premium/high engagement) vs KP281 & KP481 (moderate/low engagement).
6. Strengthen premium positioning of KP781 through performance-focused marketing and value-added features.
7. Reposition KP481 to clearly differentiate it (either enhance features or adjust pricing).
8. Target KP281 toward entry-level customers with affordability-driven campaigns and beginner-focused messaging.

▼ **Missing Value & Outlier Detection**

Missing Value Detection

1. All columns show count = 180. So, No missing values detected.
2. No nulls in categorical variables (Product, Gender, Marital Status).
3. No nulls in numerical variables (Age, Income, Usage, Miles).

4. Dataset is clean for modeling.

Outlier Detection

Miles Outliers: Extreme value around 360 miles. Several points above 200 miles (especially KP781).

Usage Outliers: Maximum usage = 7 times/week (likely KP781). Some low-usage at 2 times/week.

Income Outlier: Max income is 104,581 which is far above Q3(58,668)

Age Outlier: Max age = 50, while majority under 35.

Education max = 21 years (possible postgraduate outlier).

Distribution spread suggests KP781 has highest variance.

Missing value and outlier treatment

No missing values detected (all variables count = 180). No imputation required.

No impossible or invalid values detected. No treatment required.

Business Insights based on Non-Graphical and Visual Analysis

1. The product portfolio clearly splits into two segments: KP781 (premium, high engagement) and KP281/KP481 (moderate use mass segment).
2. KP781 customers have the highest usage, fitness levels, income, and miles, confirming strong premium positioning.
3. KP281 is the highest selling product (44%) but underperforms on engagement and income metrics.
4. KP481 shows heavy overlap with KP281 in usage, miles, and fitness, indicating weak differentiation.
5. The customer base is predominantly young (20–35 years), with peak buyers aged 24–28.
6. Sales decline significantly after age 35, and age is not a strong differentiator across products.
7. The audience is moderately male-skewed (58%) and majority partnered (59%).
8. Most customers are 14–16 years of education with middle to upper income levels.
9. KP781 has the highest variability and extreme high mile users, indicating a strong high value segment.
10. The fitness level and income are strong differentiator for product preference.

Recommendations

1. Strengthen KP781's premium positioning through performance focused branding and advanced feature messaging.
2. Clearly reposition KP481 either through feature enhancement or pricing strategy to avoid cannibalization.
3. Protect and monetize high mile KP781 users via add-ons, tiered programs, and bundled offerings.
4. Launch structured upgrade campaigns to convert KP281 users into KP781 buyers.
5. Focus marketing heavily on the 20 - 35 age segment using digital and social-first strategies.
6. Expand KP781's reach among female customers through targeted campaigns and product adaptations.
7. Target KP281 toward entry level customers with affordability driven campaigns and beginner focused messaging